



SEMESTER 2 EXAMINATIONS 2023/2024

MODULE: CA4024 - Building Complex Computational Models

PROGRAMME(S):
DS BSc in Data Science

YEAR OF STUDY: 4

EXAMINER(S):

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TIME ALLOWED: 3 Hours

INSTRUCTIONS: Answer 4 questions. All questions carry equal marks.

PLEASE DO NOT TURN OVER THIS PAGE UNTIL YOU ARE INSTRUCTED TO DO SO.

The use of programmable or text storing calculators is expressly forbidden.

Please note that where a candidate answers more than the required number of questions, the examiner will mark all questions attempted and then select the highest scoring ones.

Requirements for this paper:

- 1. Graph Paper*
- 2. Log Tables*

QUESTION 1

[TOTAL MARKS: 25]

Q 1(a)

[6 Marks]

Explain clearly, using diagrams where necessary, how continuous and discrete time series differ, giving examples of each. Explain clearly in what ways might discrete time series arise.

Q 1(b)

[6 Marks]

In the following long-run time series in Fig Q1(b), what are the principal components that can you observe? Comment on each of the figures separately.

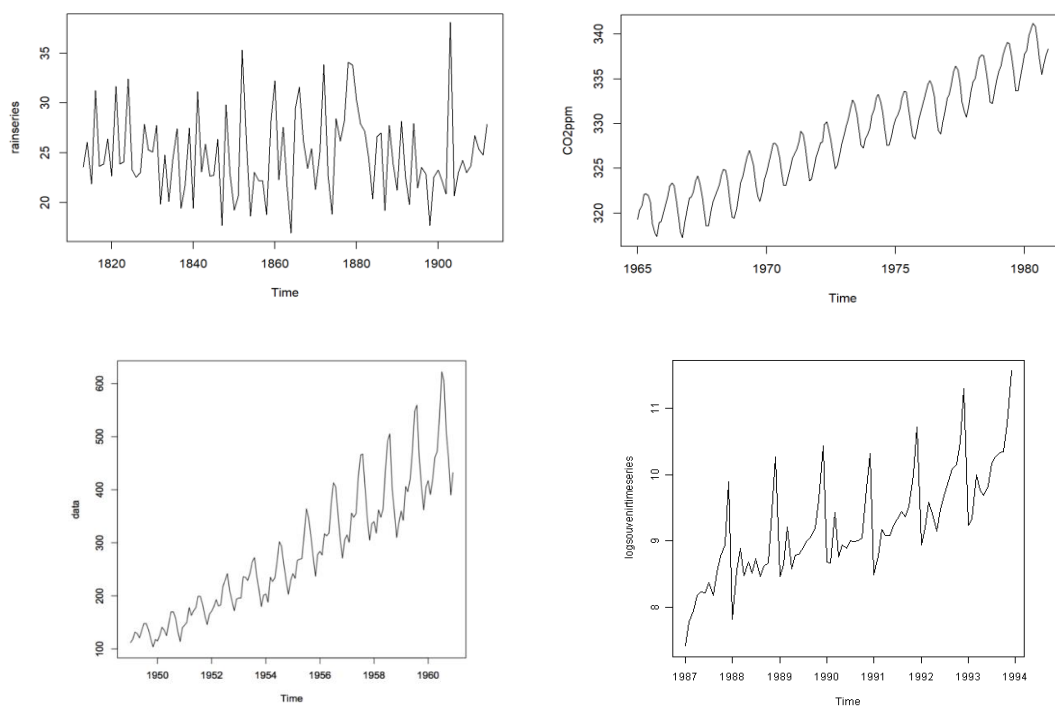


Figure Q1(b)

Q 1(c)

[6 Marks]

The following figures (Fig Q1(c)(i) and Fig Q1(c)(ii)) show ACFs and PACFs of two ARIMA models. For each, give the values of the parameters of the $ARIMA(p, d, q)$ model. Explain clearly, with reference to the ACF and PACF in both cases, your reasoning in both cases.

i. Case 1

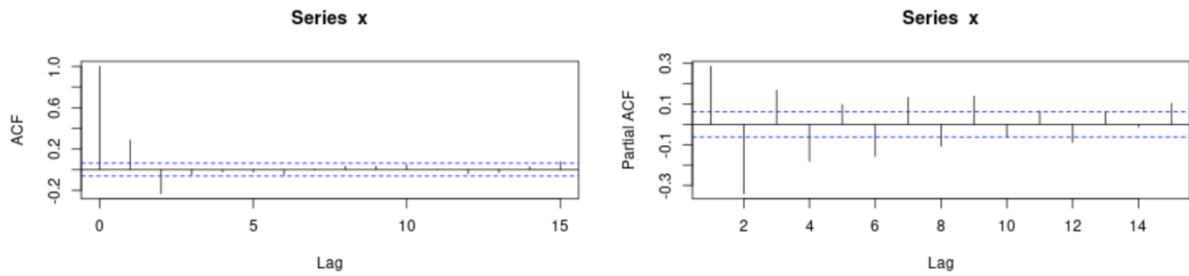


Fig Q1(c)(i)

ii. Case 2

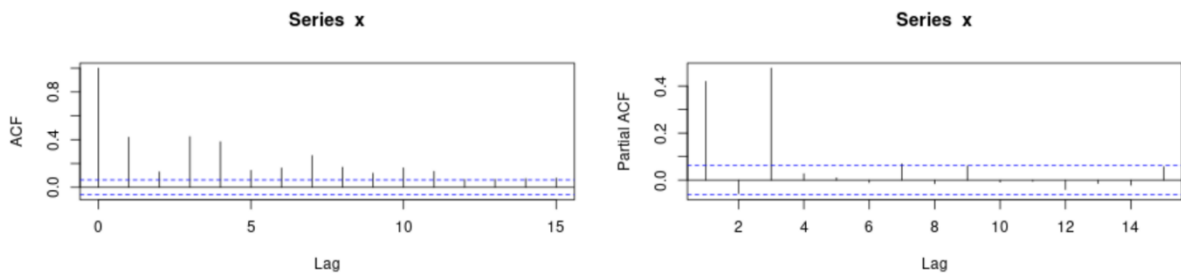


Fig Q1(c)(ii)

Q 1(d)

[7 Marks]

The figure Fig Q1(d) represents the ACF and PACF of a SARIMA model. From the diagrams point to the appropriate values of AR(1), SAR(1) and SMA(1).

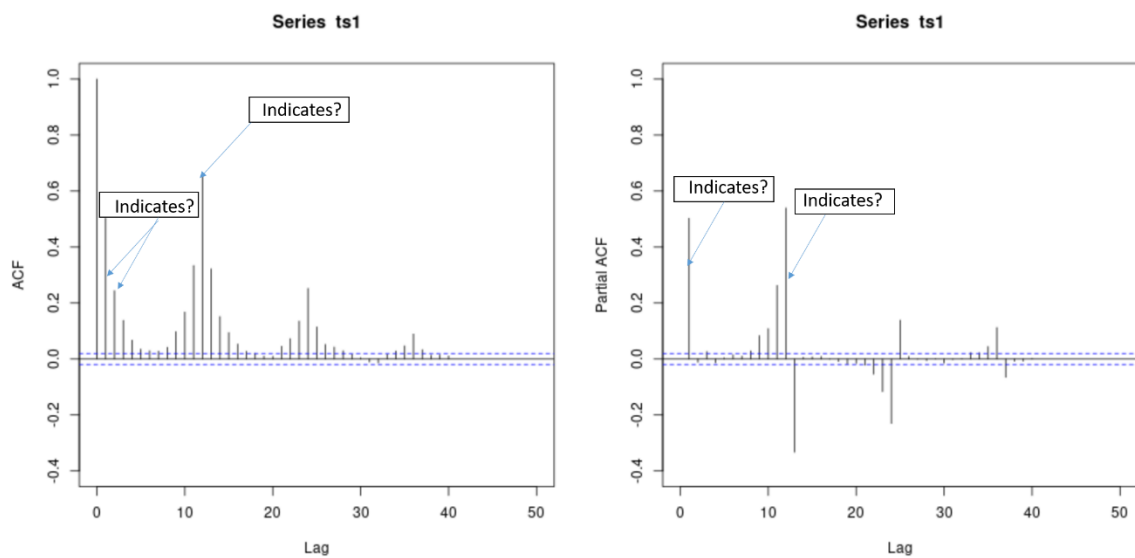


Figure Q1(d)

[End of Question 1]

QUESTION 2

[TOTAL MARKS: 25]

Q 2(a)

[4 Marks]

In the context of time series decomposition, what are the two models into which a series may be decomposed? For each, you should define them as a formula and say when they should be used.

Q 2(b)

[6 Marks]

For a time series to be decomposed into one of the two models above, define clearly what is meant by the following, saying how they might be performed on a simple time series without cyclical component: *Detrending*, *Seasonal Index*, *Seasonally Adjust*.

Q 2(c)

[15 Marks]

The time series data in Table Q2 represents quarterly values of Online orders for one retail company (in thousands) from 2015 to 2017. By graphing the data, say what kind of model should be used to decompose the data. Calculate the Trend (using 2X4 pt Moving Averages), Seasonal Index and Seasonally-Adjusted data, and draw a graph of each of them.

Year	Quarter	Online orders (000s)
2015	Q1	32
2015	Q2	22
2015	Q3	105
2015	Q4	32
2016	Q1	22
2016	Q2	25
2016	Q3	140
2016	Q4	80
2017	Q1	52
2017	Q2	38
2017	Q3	144
2017	Q4	114

Table Q2

[End of Question 2]

QUESTION 3**[TOTAL MARKS: 25]****Q 3(a)****[10 Marks]**

What are qualitative and quantitative forecasts? Give examples to show how they differ in their application.

Q 3(b)**[6 Marks]**

In forecasting, the predicted value of the dependent variable y at time t denoted $\hat{y}_t$, can be made using the regression: $\hat{y}_t = \hat{\beta}_0 + \hat{\beta}_1 x_t + \hat{\beta}_2 x_t^2 + \hat{\beta}_3 x_t^3 + \dots$. The values of the unknown coefficients $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, \hat{\beta}_3$ can either be estimated *ex post* or *ex ante*. Explain carefully what is the difference between *ex post* and *ex ante* forecasting in time series analysis.

Q 3(c)**[9 Marks]**

Explain, giving the equation, what is meant by the Naïve Forecasting Model. Does this model have any uses in forecasting practise? Show how the Simple Exponential Smoothing Model reduces to the Naïve Model in its simplest form.

[End of Question 3]**QUESTION 4****[TOTAL MARKS: 25]****Q 4(a)****[8 Marks]**

Distinguish, giving examples, between the terms emergent properties and self-organization. Is it possible for self-organization to exist without emergence?

Q 4(b)**[8 Marks]**

The following figures show the bifurcation diagram of the logistic map, r along the x -axis. Please give clear explanations for what is happening at critical points for r .

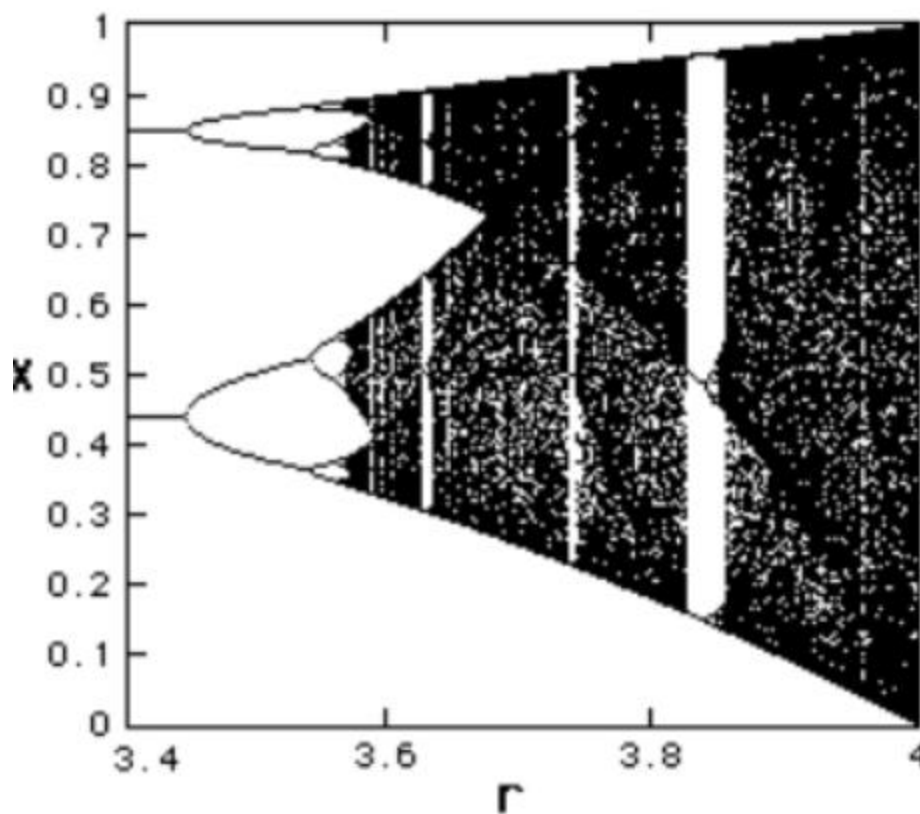


Figure Q4(b)

Q 4(c)

[7 Marks]

Describe, using an example from your experience, the steps you would take when modelling a complex system.

[End of Question 4]

QUESTION 5

[TOTAL MARKS: 25]

Q 5(a)

[4 Marks]

When does a network have a community structure? Using your own example, explain how a network can be used to model a real-life complex system and how community detection can deliver better insight into the network.

Q 5(b)

[6 Marks]

Given a weighted network graph that can be seen in Figure Q5b. Identify the adjacency matrix and calculate the modularity of the network when each node only belongs to its own community and is the only community member.

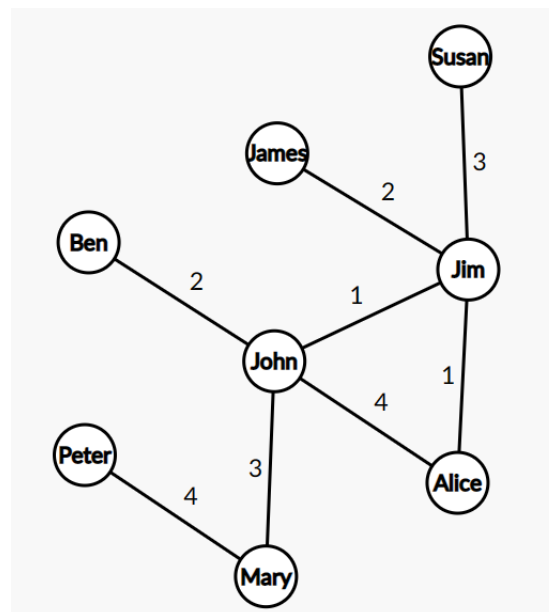


Figure Q5(b)

Q 5(c)

[10 Marks]

When detecting communities in the network in Figure Q5b above, should Alice be clustered in the same community with (Susan, Jim, James) or (John, Ben, Mary and Peter) or should Alice stay in her own community and explain why?

Q 5(d)

[5 Marks]

Figure Q5(b) above uses a network to represent the learning behaviours of students in a class where each node refers to a student and each edge between two students indicates that the two students frequently work with each other while studying.

In addition, when having a look at the students' academic performance, we found that Ben, John, Mary, Alice, Jim and Peter have passed the exam while the others failed.

Based on the community detection in Q6c and the students' academic performance, briefly discuss the relationship between the students' exam results and their learning behaviours.

[End of Question 5]

QUESTION 6

[TOTAL MARKS: 25]

Q 6(a)

[4 Marks]

Deep in the redwood forest of California, dusky-footed wood rats provide up to 80% of the diet for the spotted owl, the main predator of the wood rat.

- In the absence of wood rats, the owl population would decay by 60% each year.
- In the presence of rats, the owl population will increase each year. The owls convert rat kills into population growth with an efficiency of 50%.
- In the absence of owls, the population of the rats will grow by 10% per year.
- The owls hunt the rats, so if owls are present the rat population will be reduced by 0.1 for every owl.

If R_n , R_{n+1} , and O_n , O_{n+1} denote the rat and owl population in years n and $n+1$ and

$$R_{n+1} = aR_n + bO_n$$

$$O_{n+1} = cR_n + dO_n$$

Determine the values of a , b , c , d .

Q 6(b)

[11 Marks]

By setting up a matrix system for the above system of equations, find the maximum eigenvalue and associated eigenvector. Determine the long-term survival of the ant population.

Q 6(c)

[10 Marks]

With 30% less rats consumed per owl how will the above scenario in Q6(b) change? What is the critical rate of predation?

[End of Question 6]

[END OF EXAM]